

Navigators Technical Documentation

Navigators — Phase 38 End-to-End Test & Verification Report

Document Identifier: REP-P38-01
Classification: Quality Assurance & Engineering Verification Report
Execution Date: September 2026
Verified Result: 288 / 288 Automated Suites Passed (100% Software Suite Pass)
Physical Hardware Validation Status: BLOCKED (Pending Physical Road Testing)
Attribution: Developed by Navigators

1. Executive Summary

This report documents the comprehensive End-to-End (E2E) software and algorithmic verification conducted for Phase 38 of the Navigators platform. Testing evaluated all functional domains: - 15-state Extended Kalman Filter (EKF) covariance stability and state propagation. - Inertial Dead Reckoning with Non-Holonomic Constraints (NHC) and Zero-Velocity Updates (ZUPT). - Temporal Convolutional Network (TCN) velocity inference and ONNX WebAssembly numerical parity. - Multi-tier Role-Based Access Control (RBAC) and state machine transition rules. - Offline vector map matching, caching, and client synchronization protocols.

Final Automated Result: 288 / 288 automated test suites passed with zero regressions.

Physical Hardware Status: Physical moving vehicle road testing on a real Android smartphone under genuine tunnel satellite denial remains **BLOCKED / UNVALIDATED** until physical in-vehicle road testing is executed.

2. Test Execution Summary

Test Category	Suite Count	Passed	Failed	Blocked	Status
Automated Backend & Core Python	24	274	0	0	PASS
Automated Edge Engine (Node.js)	2	14	0	0	PASS
SUBTOTAL AUTOMATED SOFTWARE	26	288	0	0	PASS
Physical Moving Vehicle Road Test	1	0	0	1	BLOCKED
TOTAL COMBINED VERIFICATION	27	288	0	1	PARTIAL

3. Subsystem Breakdown & Evidence Log

3.1 Kinematics, Filtering & Dead Reckoning

- **tests/test_ekf.py (15/15 Passed)**: Confirms Joseph-form covariance stabilization, process noise $\mathbf{Q}_k$ scaling, and numerical positive-definiteness across 10,000 simulated steps.
- **tests/test_dead_reckoning.py (21/21 Passed)**: Validates that NHC successfully suppresses lateral velocity drift ($v_y^V \approx 0$) and vertical lift ($v_z^V \approx 0$).
- **tests/test_alignment.py (5/5 Passed)**: Asserts that Triad leveling accurately extracts the downward gravity vector and isolates forward acceleration heading within $\pm 0.5^\circ$.
- **tests/test_nav_state.py (5/5 Passed)**: Validates deterministic state machine transitions (`INIT` → `LOCKED` → `DEGRADED` → `DR` → `ZUPT` → `REACQ`).

3.2 Security, RBAC & Data Governance

- **tests/test_security_matrix.py (8/8 Passed)**: Validates route rejection (HTTP 401/403) across all 7 user roles for unauthorized direct endpoints without UI reliance.
- **tests/test_authorization_service.py (10/10 Passed)**: Confirms permission resolution, self-moderation prevention, and inactive account gating.
- **tests/test_database_rbac.py (14/14 Passed)**: Enforces foreign-key cascade deletions and junction table mappings.
- **tests/test_ownership_rules.py (2/2 Passed)**: Validates IDOR prevention; private drafts remain invisible to unauthorized accounts.
- **tests/test_auth_sessions.py (15/15 Passed)**: Asserts SHA-256 token hashing, session revocation, and expiration rules.

3.3 Data Ingestion, Models & Model Registry

- **tests/test_models.py (17/17 Passed)**: Confirms TCN causal left padding, forward tensor shapes (Batch, 2), and parameter count (5,400,322).
- **tests/test_model_registry.py (7/7 Passed)**: Enforces candidate model promotion approval gates and instant rollback execution.
- **tests/test_datasets.py (21/21 Passed)**: Validates telemetry session continuity, consent flags, and session-level splitting.

3.4 Edge Runtime & Motion Transitions (Node.js)

- **tests/vehicle_motion_detection.test.cjs (1/1 Passed)**: Verifies the complete vehicle motion lifecycle: `Stationary` → `Moving (GNSS)` → `GNSS Outage (DR)` → `Stop (ZUPT)` → `Motion Recovery`.
- **tests/offline_navigation.test.cjs (13/13 Passed)**: Validates offline vector map projection, anti-jump reacquisition capping (≤ 2.0 m per step), and causal IMU filtering.

4. Hardware Road Test: Root Cause of BLOCKED Status

[BLOCKED TEST ITEM: Physical Moving Vehicle Road Trial]

Reason: Requires physical in-vehicle driving with smartphone through real subterranean tunnel.

Current Repository State: Lab development & synthetic/benchmark replay validated.

Action Required for Unblocking: Conduct physical road test on Android device with GPS logger.

The physical road trial must remain explicitly identified as **BLOCKED**. In accordance with the project's empirical standards, automated simulation passes will never be misrepresented as physical road validation.

Developed by Navigators